

INDEFINITE THETA HESSIANS AT GREEN-FUNCTION PARAMETERS

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ABSTRACT. We construct triply connected circle domains for which the logarithmic theta Hessian at the Green-function parameter is indefinite, disproving the negative-definiteness conjecture of Saitoh in Yamada's theta-function formulation. A second-order theta expansion and explicit harmonic-measure estimates yield a positive eigenvalue of order ε^4 and a negative eigenvalue of order ε^6 .

1. INTRODUCTION

Let Ω be a bounded planar domain with $g + 1$ analytic boundary components, and let $\widehat{\Omega}$ be its Schottky double. For general background on the theta function we refer to Fay [Fay73]. We use the normalization of Yamada [Yam99]: the A -periods of the normalized holomorphic differentials are $2\pi i$, while the B -cycles are boundary curves with the orientations specified in Section 3. The B -period matrix τ is real symmetric and negative definite. Thus

$$(1) \quad \theta_\tau(z) = \sum_{n \in \mathbb{Z}^g} \exp\left(\frac{1}{2}n^T \tau n + n^T z\right).$$

The real torus relevant here is $i\mathbb{R}^g/(2\pi i\mathbb{Z}^g)$. On it, θ_τ is strictly positive. Define the *holomorphic* Hessian

$$(2) \quad H_\tau(e) = \left(\frac{\partial^2 \log \theta_\tau}{\partial z_j \partial z_k}(e)\right)_{j,k=1}^g.$$

It is a real symmetric matrix for $e \in i\mathbb{R}^g$. In particular, for $L(t) = \log \theta_\tau(it)$,

$$(3) \quad H_\tau(it) = -D_t^2 L(t).$$

For $p \in \Omega$, let D_p be the zero divisor in Ω of the differential of the Green function with pole p , with multiplicities. Its degree is g . In Abel–Jacobi notation, set

$$(4) \quad e_0(p) = D_p - p - \Delta,$$

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where Δ is the Riemann divisor class in the theta convention of [Fay73, Yam99]. Here a divisor class means divisors modulo linear equivalence, and Δ has degree $g - 1$; it is the distinguished class associated with the chosen theta function, satisfying $2\Delta = K_{\widehat{\Omega}}$ as divisor classes. Thus $D_p - p - \Delta$ has degree $g - 1 - (g - 1) = 0$ and defines a point of the Jacobian. The displayed subtraction is a subtraction of divisor classes followed by the Abel–Jacobi identification, rather than a subtraction of points in the plane. A representative belongs to $i\mathbb{R}^g$, and the Hessian is independent of the allowed period ambiguity. The definiteness problem considered here originates in Saitoh’s work on reproducing kernels. In [Sai79, pp. 276–277, equation (7.11)], Saitoh proved a positive-kernel comparison between a product of two Rudin kernels and the Bergman kernel in the doubly connected case, and asked whether it extends to arbitrary compact bordered Riemann surfaces. This is a comparison in the order of positive definite kernels, stronger than its diagonal consequence in equation (7.12) of that paper. It should also be distinguished from the comparison with the *exact* Bergman kernel proved earlier in [Sai77, Corollary 3.1].

In the theta-function formulation, the proposed conclusion is

$$(5) \quad H_{\tau}(e_0(p)) < 0.$$

Yamada explicitly attributes this negative-definiteness problem to Saitoh in [Yam99, p. 1399], citing [Sai88, p. 37]; our attribution to that page of Saitoh’s book follows Yamada’s account. The formulation also appears in [Yam98, p. 41]. We therefore refer to (5) as Saitoh’s negative-definiteness conjecture in Yamada’s theta-function formulation. The scalar comparison between the conjugate Hardy kernel and the Bergman kernel was subsequently proved by Guan [Gua19]. The matrix assertion additionally requires the quadratic form (2) to be negative in every nonzero real direction.

The purpose of this paper is to disprove (5). The first case in which a transverse direction is available is $g = 2$, and our examples have precisely this genus.

Fix $0 < \rho < 1$ and put

$$(6) \quad L = \log(1/\rho), \quad d_* = \frac{2\pi^2}{L}.$$

For small $\varepsilon > 0$, let $c = 1 - \varepsilon$ and define

$$(7) \quad K_{\pm}(c) = \left\{ z \in \mathbb{D} : \left| \frac{z \mp c}{1 \mp cz} \right| \leq \rho \right\},$$

$$(8) \quad \Omega_{\varepsilon} = \mathbb{D} \setminus (K_+(c) \cup K_-(c)), \quad p_{\varepsilon} = i(1 - \varepsilon^5).$$

For $0 < \varepsilon < 1 - \rho$, these are disjoint closed Euclidean disks strictly inside $\mathbb{D}$, so Ω_{ε} is a regular triply connected domain. The centers and the common radius of the disks are, respectively,

$$(9) \quad \pm \frac{c(1 - \rho^2)}{1 - c^2\rho^2}, \quad \frac{\rho(1 - c^2)}{1 - c^2\rho^2}.$$

Indeed, the positive center in (9) exceeds the radius exactly when $c > \rho$, since

$$c(1 - \rho^2) - \rho(1 - c^2) = (c - \rho)(1 + c\rho) > 0.$$

Thus neither disk meets the imaginary axis, and $p_\varepsilon \in \Omega_\varepsilon$. Figure 1 illustrates the geometry of this family.

Write τ_ε for the period matrix of the double of this actual domain, in the normalization just specified. It is determined by Ω_ε , rather than prescribed as a separate parameter. Sections 3 and 4 will prove the exact identity

$$\tau_\varepsilon = -\pi C_\varepsilon = \begin{pmatrix} -d_\varepsilon & b_\varepsilon \\ b_\varepsilon & -d_\varepsilon \end{pmatrix},$$

$$(C_\varepsilon)_{jk} = \int_{\Omega_\varepsilon} \nabla \omega_j \cdot \nabla \omega_k dA, \quad j, k \in \{+, -\},$$

where $\omega_\pm$ are the harmonic measures of the two holes. In particular, the quantities $d_\varepsilon, b_\varepsilon$ used in the theta calculation below are ultimately obtained from the geometry.

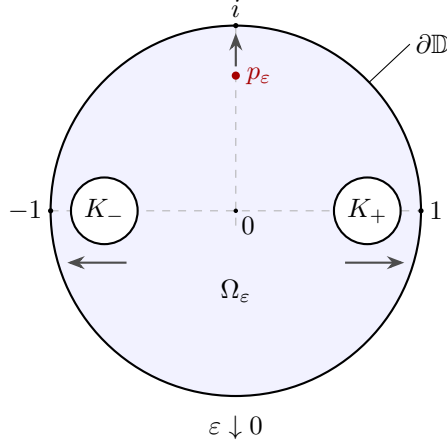


FIGURE 1. The shaded domain Ω_ε is the unit disk with the two closed disks $K_\pm(c)$ removed. As $\varepsilon \downarrow 0$, the holes shrink towards ± 1 and the pole $p_\varepsilon = i(1 - \varepsilon^5)$ tends to i . The diagram is schematic; the pole-to-boundary distance is enlarged for visibility.

For $d > 0$, write

$$(10) \quad P_d(t) = \sum_{n \in \mathbb{Z}} e^{-dn^2/2 + int}, \quad \ell_d(t) = \log P_d(t).$$

Lemma 2.1 proves that there is a unique $t_d \in (0, \pi)$ such that $\ell_d''(t_d) = 0$, and that

$$(11) \quad \kappa_d = \ell_d'''(t_d) > 0.$$

Let $\kappa_* := \kappa_{d_*}$.

Theorem 1.1. *Fix $0 < \rho < 1$ and let $\Omega_\varepsilon, p_\varepsilon$ be given by (8). In the normalized basis associated with the two inner boundary components, the matrix*

$$H_{\tau_\varepsilon}(e_0(p_\varepsilon))$$

has eigenvectors $(1, 1)$ and $(1, -1)$. Their respective eigenvalues satisfy

$$(12) \quad \lambda_{\parallel}(\varepsilon) = -\frac{\pi\kappa_*}{L}\varepsilon^6 + O(\varepsilon^7),$$

$$(13) \quad \lambda_{\perp}(\varepsilon) = \frac{\pi^4\kappa_*^2}{L^4}\varepsilon^4 + O(\varepsilon^5).$$

Consequently, for all sufficiently small positive ε , the matrix has one strictly positive and one strictly negative eigenvalue.

For each sufficiently small positive ε , all three boundary components are nondegenerate and the double is a smooth genus-two surface. In particular, the indefiniteness occurs on genuine Jacobians of planar doubles. The theorem answers the proposed universal negative-definiteness assertion; it does not classify the signature of $H_\tau(e)$ for arbitrary parameters e .

The mechanism can be described in terms of two small parameters. The off-diagonal coupling b in the period matrix tends to zero, so the theta function tends to a product of elliptic theta functions. At an auxiliary extremum, stationarity forces one Hessian eigenvalue to vanish. A second-order expansion gives the other eigenvalue as $b^2\kappa^2 + O(b^3) > 0$. The Abel displacement δ associated with an interior pole then moves the extremum and makes the vanishing eigenvalue negative at order δ . The geometry is chosen so that $\delta = o(b^2)$, preserving the positive eigenvalue.

The proof has three parts. Section 2 isolates the positive transverse term in a theta expansion. Sections 3 and 4 realize the theta parameters by the circle domains and estimate their sizes. Section 5 identifies the maximizer with (4) and proves the theorem. Appendix A gives numerical illustrations and formulas for reproducing them.

2. A PERTURBATION OF A PRODUCT THETA FUNCTION

2.1. A nondegenerate one-dimensional maximum.

Lemma 2.1. *For every $d > 0$, the function $-\ell'_d$ on $\mathbb{R}/(2\pi\mathbb{Z})$ has a unique maximum at a point $t_d \in (\pi/2, \pi)$. This maximum is nondegenerate. More precisely, $\ell''_d(t_d) = 0$ and $\kappa_d > 0$. The functions $d \mapsto t_d$ and $d \mapsto \kappa_d$ are smooth.*

Proof. Set $h_d = -\ell''_d$ and $q = e^{-d/2}$. The Jacobi product formula is

$$P_d(t) = \prod_{m \geq 1} (1 - q^{2m})(1 + 2q^{2m-1} \cos t + q^{4m-2}).$$

Every factor is positive for real t . For $f_r(t) = 1 + 2r \cos t + r^2$ one has

$$-\frac{d^2}{dt^2} \log f_r(t) = \frac{2r \cos t}{f_r(t)} + \frac{4r^2 \sin^2 t}{f_r(t)^2} = \frac{2r((1 + r^2) \cos t + 2r)}{f_r(t)^2}.$$

With $r_m = q^{2m-1}$, summing these logarithmic derivatives gives

$$(14) \quad h_d(t) = 2 \sum_{m \geq 1} \frac{r_m((1 + r_m^2) \cos t + 2r_m)}{(1 + 2r_m \cos t + r_m^2)^2}.$$

The product and its real derivatives converge uniformly. In particular,

$$\begin{aligned} h_d(0) &= 2 \sum_{m \geq 1} \frac{r_m}{(1 + r_m)^2} > 0, \\ h_d(\pi) &= -2 \sum_{m \geq 1} \frac{r_m}{(1 - r_m)^2} < 0, \\ h_d(\pi/2) &= 4 \sum_{m \geq 1} \frac{r_m^2}{(1 + r_m^2)^2} > 0. \end{aligned}$$

For completeness, strict monotonicity on $(0, \pi)$ can be read off from the elliptic extension. The entire function P_d has simple zeros at

$$\pi + id/2 + 2\pi\mathbb{Z} + id\mathbb{Z},$$

and satisfies $P_d(t + id) = e^{d/2 - it} P_d(t)$. Indeed, completing the square in the defining series and replacing $n + 1$ by a new index gives

$$\sum_{n \in \mathbb{Z}} e^{-dn^2/2 - dn + int} = e^{d/2 - it} \sum_{n \in \mathbb{Z}} e^{-d(n+1)^2/2 + i(n+1)t}.$$

Taking two derivatives of the logarithm removes the affine logarithm of the multiplier, so that $(\log P_d)''$ is periodic. Hence h_d , extended meromorphically, has periods $2\pi, id$ and one double pole per fundamental parallelogram, with principal part $(t - \pi - id/2)^{-2}$. Subtracting the Weierstrass function with this pole and these periods leaves an entire elliptic function. Such a function is bounded on a period parallelogram and hence constant. Therefore

$$(15) \quad h_d(t) = \wp(t - \pi - id/2; 2\pi\mathbb{Z} + id\mathbb{Z}) + C_d.$$

This also follows from the standard theta–Weierstrass identities [DLMF, §23.6]. The zeros of $\wp'$ are exactly the three nonzero half-periods. The segment $t - \pi - id/2$, $0 < t < \pi$, contains no half-period and no pole. Thus h'_d has no zero there. It is real and continuous, and the two endpoint values show that it is strictly negative.

It follows that h_d has exactly one zero t_d in $(0, \pi)$, necessarily in $(\pi/2, \pi)$. Evenness and periodicity give its only other zero on the real period circle, namely $2\pi - t_d$. Since $(-\ell'_d)' = h_d$, the first zero is the unique maximum of $-\ell'_d$ and the other is its unique minimum. Moreover $\ell_d'''(t_d) = -h'_d(t_d) > 0$. Smooth dependence follows from the implicit function theorem, since this derivative does not vanish. $\square$

2.2. Uniform analytic expansion in the coupling. For $|b| < d$ define

$$(16) \quad \Theta_{d,b}(z_1, z_2) = \sum_{m,n \in \mathbb{Z}} e^{-\frac{d}{2}(m^2+n^2)+bmn+mz_1+nz_2},$$

$$(17) \quad \mathcal{L}_{d,b}(t_1, t_2) = \log \Theta_{d,b}(it_1, it_2), \quad \mathcal{H}_{d,b}(it) = -D_t^2 \mathcal{L}_{d,b}(t).$$

All variables t_j are understood modulo 2π . Set

$$A = \begin{pmatrix} d & -b \\ -b & d \end{pmatrix} > 0.$$

To fix the constant in Poisson summation, use the Fourier transform

$$\widehat{f}(\xi) = \int_{\mathbb{R}^2} f(x) e^{-ix \cdot \xi} dx, \quad f_t(x) = e^{-x^T A x / 2 + it \cdot x}.$$

Diagonalize A and apply the Gaussian integral in each coordinate to obtain

$$\widehat{f}_t(\xi) = \frac{2\pi}{\sqrt{\det A}} \exp\left(-\frac{1}{2}(\xi - t)^T A^{-1}(\xi - t)\right).$$

For this Fourier convention the Poisson formula is

$$\sum_{n \in \mathbb{Z}^2} f_t(n) = \sum_{k \in \mathbb{Z}^2} \widehat{f}_t(2\pi k).$$

Since f_t is a Schwartz function, both sides converge absolutely. Substitution gives

$$(18) \quad \Theta_{d,b}(it) = \frac{2\pi}{\sqrt{\det A}} \sum_{k \in \mathbb{Z}^2} \exp\left(-\frac{1}{2}(t - 2\pi k)^T A^{-1}(t - 2\pi k)\right) > 0.$$

Thus the real logarithm in (17) is defined everywhere on the real torus.

In the remainder of this section d lies in a sufficiently small compact neighborhood $I \Subset (0, \infty)$ of an arbitrary fixed $d_* > 0$. Constants in all remainder estimates can be chosen uniformly for $d \in I$.

Lemma 2.2. *Put $\ell = \ell_d$, and abbreviate $p_j = \ell'(t_j)$, $q_j = \ell''(t_j)$. Then, as $b \rightarrow 0$,*

$$(19) \quad \begin{aligned} \mathcal{L}_{d,b}(t_1, t_2) &= \ell(t_1) + \ell(t_2) - bp_1 p_2 \\ &\quad + \frac{b^2}{2}(q_1 q_2 + q_1 p_2^2 + p_1^2 q_2) + O(b^3). \end{aligned}$$

The expansion holds in every fixed C^r norm on the real torus, uniformly for $d \in I$.

Proof. For $|b| \leq b_0 < \min I$, the quadratic exponent is bounded above by $-(\min I - b_0)(m^2 + n^2)/2$. Differentiating a summand in b , in d , or in t only introduces polynomial factors. All such differentiated series converge uniformly by Gaussian summability. More explicitly, the scalar Taylor remainder obeys

$$\left| e^{bmn} - 1 - bmn - \frac{b^2 m^2 n^2}{2} \right| \leq \frac{|b|^3 |mn|^3}{6} e^{b_0 |mn|}.$$

Multiplying by the uncoupled Gaussian, and using $2|mn| \leq m^2 + n^2$, gives a summable Gaussian times a polynomial. Applying $\partial_1^a \partial_2^c$ only adds factors of absolute value $|m|^a |n|^c$. This proves a bound by a constant times $|b|^3$ in each prescribed C^r norm, and justifies all the subsequent termwise differentiations.

Let $P = P_d$ and $U(t_1, t_2) = P(t_1)P(t_2)$. Since differentiating twice, once in each real variable, multiplies a Fourier mode by $-mn$, expansion of e^{bmn} gives

$$\Theta_{d,b}(it) = U - b \partial_1 \partial_2 U + \frac{b^2}{2} \partial_1^2 \partial_2^2 U + O(b^3).$$

Dividing by U and using $P'/P = \ell'$ and $P''/P = \ell'' + (\ell')^2$, we obtain

$$\frac{\Theta_{d,b}(it)}{U} = 1 - bp_1 p_2 + \frac{b^2}{2} (q_1 + p_1^2)(q_2 + p_2^2) + O(b^3).$$

If the last display is written as $1 + bx + b^2 y + O(b^3)$, then

$$\log(1 + bx + b^2 y + O(b^3)) = bx + b^2 \left(y - \frac{x^2}{2} \right) + O(b^3), \quad x = -p_1 p_2.$$

Consequently

$$y - \frac{x^2}{2} = \frac{1}{2} [(q_1 + p_1^2)(q_2 + p_2^2) - p_1^2 p_2^2] = \frac{1}{2} (q_1 q_2 + q_1 p_2^2 + p_1^2 q_2).$$

The term $-\frac{1}{2} b^2 p_1^2 p_2^2$ in the logarithm thus cancels the corresponding fourth-degree product. This proves (19). A uniform positive lower bound for U , and then for $\Theta_{d,b}(it)$ for small b , follows from positivity and compactness. Thus taking logarithms and derivatives preserves the uniform remainder bounds. $\square$

2.3. A positive transverse eigenvalue. Let $D = \partial_1 + \partial_2$ and define

$$(20) \quad S_{d,b}(t) = -D \mathcal{L}_{d,b}(t).$$

Proposition 2.3. *For sufficiently small $|b|$, $S_{d,b}$ has a unique global maximum on the real torus. It is nondegenerate and has the form $(u_{d,b}, u_{d,b})$, where $u_{d,b} = t_d + O(b)$. At this point the eigenvalues of $\mathcal{H}_{d,b}$ in the directions $(1, 1)$ and $(1, -1)$ are, respectively,*

$$(21) \quad 0, \quad b^2 \kappa_d^2 + O(b^3).$$

Moreover, if $\Lambda_{\parallel}(u)$ is the $(1, 1)$ eigenvalue of $\mathcal{H}_{d,b}(iu, iu)$, then

$$(22) \quad \Lambda'_{\parallel}(u_{d,b}) = -\kappa_d + O(b).$$

Proof. At $b = 0$ the function in (20) is $-\ell'_d(t_1) - \ell'_d(t_2)$. By Lemma 2.1, its unique global maximum is (t_d, t_d) and its Hessian there is $-\kappa_d I_2$. For the precise use of the implicit function theorem, set

$$F(t, b) = \nabla_t S_{d,b}(t), \quad t^0 = (t_d, t_d).$$

Then $F(t^0, 0) = 0$ and $D_t F(t^0, 0) = D_t^2 S_{d,0}(t^0) = -\kappa_d I_2$ is invertible. The theorem therefore solves the two stationarity equations $F(t, b) = 0$ for a

unique smooth branch $t = t(b)$ near t^0 . Smoothness gives $t(b) = t^0 + O(b)$, and continuity of the Hessian makes this critical point a strict local maximum. To verify global uniqueness, choose a small convex coordinate neighborhood on which the Hessian at $b = 0$ is uniformly negative definite. Outside this neighborhood the maximum has a strictly positive value gap, by compactness. Both the gap and strict concavity persist under a sufficiently small C^2 perturbation. These choices can be made uniformly for $d \in I$, after shrinking I if necessary.

The identity $S_{d,b}(t_1, t_2) = S_{d,b}(t_2, t_1)$ implies that exchanging the coordinates of a maximum produces another maximum. Uniqueness forces the two points to coincide. In the chosen coordinate neighborhood this means $t_1 = t_2 = u$, with $u = t_d + O(b)$.

Write $\mathcal{L} = \mathcal{L}_{d,b}$, with subscripts denoting real partial derivatives. It is $S = -\mathcal{L}_1 - \mathcal{L}_2$, not $\mathcal{L}$, that is stationary here. Thus

$$0 = \partial_1 S(u, u) = -(\mathcal{L}_{11} + \mathcal{L}_{12})(u, u), \quad 0 = \partial_2 S(u, u) = -(\mathcal{L}_{21} + \mathcal{L}_{22})(u, u).$$

The first stationarity equation is

$$(23) \quad \mathcal{L}_{11}(u, u) + \mathcal{L}_{12}(u, u) = 0.$$

Set $p = \ell'_d(u)$, $q = \ell''_d(u)$ and $r = \ell'''_d(u)$. Then $q = O(b)$ and $r = \kappa_d + O(b)$. Here $q = O(b)$ follows by Taylor expansion of ℓ''_d at t_d , where it vanishes; also $p = O(1)$ uniformly. To take the mixed derivative in (19), put $r_j = \ell'''_d(t_j)$ and note separately that

$$\begin{aligned} \partial_1 \partial_2(p_1 p_2) &= q_1 q_2, & \partial_1 \partial_2(q_1 q_2) &= r_1 r_2, \\ \partial_1 \partial_2(q_1 p_2^2) &= 2r_1 p_2 q_2, & \partial_1 \partial_2(p_1^2 q_2) &= 2p_1 q_1 r_2. \end{aligned}$$

Evaluating at (u, u) gives the full second-order expression

$$(24) \quad \mathcal{L}_{12}(u, u) = -bq^2 + \frac{b^2}{2}(r^2 + 4pqr) + O(b^3) = \frac{b^2}{2}\kappa_d^2 + O(b^3).$$

Since $q = O(b)$ at the perturbed maximum, the contribution $-bq^2$ is of order b^3 ; the strictly positive term $b^2\kappa_d^2/2$ determines the leading sign.

Exchange symmetry of $\mathcal{L}$ gives $\mathcal{L}_{11}(s, s) = \mathcal{L}_{22}(s, s)$ for every real s . Therefore, on the entire diagonal, the matrix has the form

$$\mathcal{H}_{d,b}(is, is) = \begin{pmatrix} a(s) & c(s) \\ c(s) & a(s) \end{pmatrix}, \quad a(s) = -\mathcal{L}_{11}(s, s), \quad c(s) = -\mathcal{L}_{12}(s, s).$$

Direct multiplication gives

$$\begin{pmatrix} a & c \\ c & a \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = (a + c) \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad \begin{pmatrix} a & c \\ c & a \end{pmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = (a - c) \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

Thus both eigenvectors are fixed along the diagonal. At $s = u$, the additional stationarity relation (23) says $a + c = 0$. The other eigenvalue is then $a - c = -2c = 2\mathcal{L}_{12}$, which proves (21). Finally,

$$\Lambda_{\parallel}(u) = -(\mathcal{L}_{11} + \mathcal{L}_{12})(u, u).$$

At $b = 0$ this equals $-\ell_d''(u)$, whose derivative at t_d is $-\kappa_d$. Smooth dependence and $u_{d,b} = t_d + O(b)$ prove (22). $\square$

2.4. Moving the maximum into a finite theta ratio. For $\delta > 0$ set

$$(25) \quad R_{d,b,\delta}(t) = \mathcal{L}_{d,b}(t - \delta \mathbf{1}) - \mathcal{L}_{d,b}(t), \quad \mathbf{1} = (1, 1).$$

Proposition 2.4. *For $d \in I$ and sufficiently small $|b|, \delta > 0$, (25) has a unique global maximum, at a point $v_{d,b,\delta} \mathbf{1}$. It satisfies*

$$(26) \quad v_{d,b,\delta} = u_{d,b} + \frac{\delta}{2} + O(\delta^2).$$

At this maximum the two eigenvalues of $\mathcal{H}_{d,b}$ satisfy

$$(27) \quad \lambda_{\parallel}(d, b, \delta) = -\frac{\kappa_d}{2}\delta + O(|b|\delta + \delta^2),$$

$$(28) \quad \lambda_{\perp}(d, b, \delta) = b^2 \kappa_d^2 + O(|b|^3 + \delta).$$

All remainders are uniform for the indicated parameters.

Proof. Set $t = w + \delta \mathbf{1}/2$, so the two arguments of $\mathcal{L}$ in the ratio are $w - \delta \mathbf{1}/2$ and $w + \delta \mathbf{1}/2$. If $f(s) = \mathcal{L}(w + s \mathbf{1})$, then $f^{(j)}(0) = D^j \mathcal{L}(w)$. Subtracting the Taylor expansions at $s = \pm \delta/2$ cancels all even powers:

$$f(-\delta/2) - f(\delta/2) = -\delta f'(0) - \frac{\delta^3}{24} f'''(0) + O(\delta^5).$$

The coefficient $1/24$ is $2/(2^3 3!)$. Dividing by δ yields, in every fixed C^r norm,

$$(29) \quad \frac{R_{d,b,\delta}(w + \frac{\delta}{2} \mathbf{1})}{\delta} = -D \mathcal{L}_{d,b}(w) - \frac{\delta^2}{24} D^3 \mathcal{L}_{d,b}(w) + O(\delta^4).$$

Denote the left side by $Q_{\delta}(w)$ and extend it at $\delta = 0$ by $Q_0 = S_{d,b}$. Proposition 2.3 gives the unique nondegenerate maximum $w_0 = u_{d,b} \mathbf{1}$ of Q_0 . The compact value gap and local strict concavity persist because $Q_{\delta} - S_{d,b} = O(\delta^2)$ in C^2 . Hence Q_{δ} has a unique global maximum w_{δ} near w_0 .

To make its displacement estimate explicit, write $h = w_{\delta} - w_0$. Since both gradients vanish at their respective maxima,

$$\left(\int_0^1 D^2 S_{d,b}(w_0 + sh) ds \right) h = \nabla S_{d,b}(w_{\delta}) - \nabla Q_{\delta}(w_{\delta}) = O(\delta^2).$$

The matrix in parentheses is uniformly negative definite in the chosen neighborhood, so its inverse is uniformly bounded. Consequently $h = O(\delta^2)$.

Exchange symmetry of Q_{δ} , together with uniqueness, now gives $w_{\delta} = (s_{\delta}, s_{\delta})$, where $s_{\delta} = u_{d,b} + O(\delta^2)$. Returning to the original variable gives

$$t_{\delta} = w_{\delta} + \frac{\delta}{2} \mathbf{1} = \left(u_{d,b} + \frac{\delta}{2} + O(\delta^2) \right) \mathbf{1}.$$

This proves (26). Symmetry supplies equality of the coordinates; the centered expansion and the inverse Hessian estimate supply the displacement and its error.

The moving point stays on the diagonal, so the matrix form established in Proposition 2.3 applies at both $u_{d,b}\mathbf{1}$ and $v_{d,b,\delta}\mathbf{1}$. The eigenvectors $(1, 1)$ and $(1, -1)$ therefore do not move, although their eigenvalues do. With $u = u_{d,b}$ and $v = v_{d,b,\delta}$, Taylor expansion gives

$$\begin{aligned}\lambda_{\parallel}(d, b, \delta) &= \Lambda_{\parallel}(u) + \Lambda'_{\parallel}(u)(v - u) + O((v - u)^2) \\ &= 0 + (-\kappa_d + O(b)) \left(\frac{\delta}{2} + O(\delta^2) \right) + O(\delta^2) \\ &= -\frac{\kappa_d}{2}\delta + O(|b|\delta + \delta^2).\end{aligned}$$

For the other fixed direction, put $\Lambda_{\perp}(s) = -(\mathcal{L}_{11} - \mathcal{L}_{12})(s, s)$. Its derivative is uniformly bounded, and hence

$$\Lambda_{\perp}(v) = \Lambda_{\perp}(u) + O(|v - u|) = b^2\kappa_d^2 + O(|b|^3) + O(\delta).$$

These are (27) and (28). The zero longitudinal eigenvalue belongs only to the auxiliary point $u\mathbf{1}$; it is not asserted at the displaced maximum. $\square$

Remark 2.5. The negative eigenvalue has no error term independent of δ . This is a consequence of the exact row-null relation (23). It permits the choice $\delta \ll b^2$. A bound of the form $-\kappa_d\delta/2 + O(b^2 + \delta^2)$ would be insufficient for the construction.

Figure 2 summarizes the local sign change. Here $\Lambda_{\perp}(t)$ denotes the transverse eigenvalue of $\mathcal{H}_{d,b}(it, it)$, alongside $\Lambda_{\parallel}(t)$.

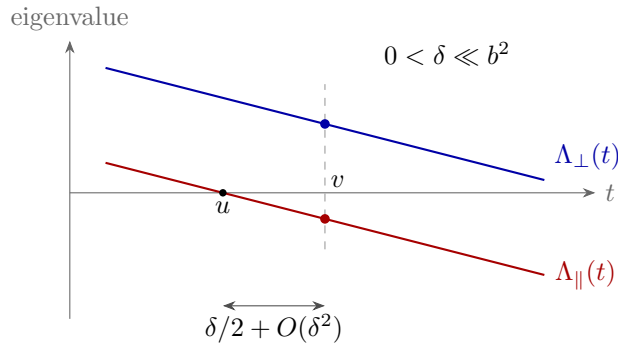


FIGURE 2. Local eigenvalue behavior for fixed small $b > 0$, with $u = u_{d,b}$ and $v = v_{d,b,\delta}$. The longitudinal eigenvalue vanishes at u and becomes negative at v , while the transverse eigenvalue remains positive. Straight segments indicate the local behavior only; the horizontal and vertical scales are schematic.

3. PERIOD NORMALIZATION FROM HARMONIC MEASURES

We record the relation between periods and the planar Dirichlet problem with orientations, in order to fix the signs and factors in the construction.

Let Γ_0 be the outer boundary of a regular $(g+1)$ -connected planar domain, and let $\Gamma_1, \dots, \Gamma_g$ be the inner boundaries. Let ω_j be the harmonic function with boundary values 1 on Γ_j and 0 on the other components. Define

$$(30) \quad C_{jk} = \int_{\Omega} \nabla \omega_j \cdot \nabla \omega_k dA = \int_{\Gamma_j} \partial_{\nu} \omega_k ds,$$

where ν is the outward unit normal of Ω , pointing into a hole on an inner boundary. For clarity, Green's first identity gives

$$\begin{aligned} \int_{\Omega} \nabla \omega_j \cdot \nabla \omega_k dA &= \int_{\partial\Omega} \omega_j \partial_{\nu} \omega_k ds - \int_{\Omega} \omega_j \Delta \omega_k dA \\ &= \sum_{\ell=0}^g \int_{\Gamma_{\ell}} \omega_j \partial_{\nu} \omega_k ds = \int_{\Gamma_j} \partial_{\nu} \omega_k ds. \end{aligned}$$

The interior term vanishes by harmonicity; all boundary terms except $\ell = j$ vanish by the boundary values of ω_j .

Symmetry of C follows from the energy expression. To check strict positive definiteness, let $a \in \mathbb{R}^g$ and set $v = \sum_j a_j \omega_j$. Then

$$a^T C a = \int_{\Omega} \left| \sum_j a_j \nabla \omega_j \right|^2 dA = \int_{\Omega} |\nabla v|^2 dA \geq 0.$$

If this is zero, v is constant on the connected domain. Since $v = 0$ on Γ_0 , that constant is zero. But $v = a_j$ on Γ_j , so all $a_j = 0$. This proves $C > 0$.

An integral $\int_{\Gamma} \partial_{\nu} v ds$ will be called the *normal flux* of v through Γ . It measures the integrated normal derivative of a potential; it is not a winding number.

Use the Hodge star $*dx = dy$, $*dy = -dx$, and set

$$(31) \quad u_k = -i\pi(d\omega_k + i * d\omega_k).$$

In local coordinates $z = x + iy$,

$$d\omega_k + i * d\omega_k = ((\omega_k)_x - i(\omega_k)_y)(dx + i dy) = 2\partial\omega_k.$$

The coefficient is holomorphic because $\partial_{\bar{z}}((\omega_k)_x - i(\omega_k)_y) = \Delta\omega_k/2 = 0$. Also $u_k = \pi*d\omega_k - i\pi d\omega_k$, so the imaginary part of its integral is determined by the change in $-\pi\omega_k$. Its multiple u_k is real on tangent vectors to every boundary, because $d\omega_k$ vanishes on those vectors. It therefore extends by Schwarz reflection to a holomorphic differential on the double, with $\phi^* u_k = \overline{u_k}$.

Orient B_j counterclockwise around the j th hole. Let A_j run in the front copy from Γ_j to Γ_0 and return along the reflected path in the back copy. These paths can be chosen as the usual disjoint crosscuts; with these orientations they form a canonical basis. If I is the integral of u_k along the front

path, the full A_j -integral is $I - \bar{I}$. Along the front path the change in ω_k is $0 - \delta_{jk}$, and therefore

$$\operatorname{Im} I = -\pi(0 - \delta_{jk}) = \pi\delta_{jk}, \quad I - \bar{I} = 2i \operatorname{Im} I.$$

This gives

$$(32) \quad \int_{A_j} u_k = 2\pi i \delta_{jk}.$$

Furthermore, $\int_{B_j} d\omega_k = 0$, and $\int_{B_j} *d\omega_k$ is the flux with normal pointing out of the hole, the negative of the normal in (30). One can see the sign directly: if $T = (T_x, T_y)$ is the unit tangent to the counterclockwise circle, its right normal is $(T_y, -T_x)$ and points out of the hole. Then

$$(*d\omega_k)(T) = (\omega_k)_x T_y - (\omega_k)_y T_x = \nabla \omega_k \cdot (T_y, -T_x) = -\partial_\nu \omega_k.$$

Since the real part of u_k is $\pi * d\omega_k$, integration gives

$$(33) \quad \tau_{jk} = \int_{B_j} u_k = -\pi C_{jk}.$$

In particular, the geometric period matrix is real symmetric and strictly negative definite because $C > 0$. The cycle orientations and the two opposite normals are illustrated in Figure 3.

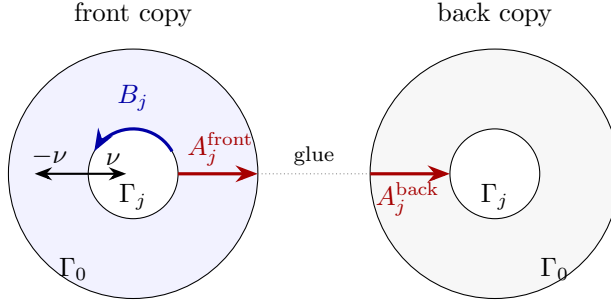


FIGURE 3. A schematic for one inner boundary component; other holes are omitted. Corresponding inner and outer boundaries are identified in the double. The cycle A_j goes from Γ_j to Γ_0 in front and returns in back. In front, B_j is counterclockwise. Its flux normal $-\nu$ points out of the hole, whereas the outward normal ν of the domain points into the hole.

Lemma 3.1. *Evaluate $p - \phi(p)$ along a symmetric path from $\phi(p)$ to p passing through Γ_0 . Then, in the above normalized coordinates,*

$$(34) \quad p - \phi(p) = -2\pi i (\omega_1(p), \dots, \omega_g(p)).$$

Proof. Let J_k be the integral of u_k on the front half of the path, from Γ_0 to p . The full integral is $J_k - \overline{J_k}$, and $\text{Im } J_k = -\pi\omega_k(p)$. This is (34). Changing the admissible paths by A -periods only adds $2\pi i\mathbb{Z}^g$, which does not affect theta values or their Hessians. $\square$

Remark 3.2 (An annulus check). For the annulus $\rho < |z| < 1$, the harmonic measure is $\log |z| / \log \rho$ and its energy is $2\pi / \log(1/\rho)$. Formula (33) gives the scalar period $-2\pi^2 / \log(1/\rho)$. This verifies the same factor d_* as in (6).

4. EXPLICIT ESTIMATES FOR THE CIRCLE DOMAINS

In this section $c \rightarrow 1^-$ and ρ is fixed. Let Ω_c be the domain in (8), and label the inner harmonic measures by ω_+, ω_- . We suppress c from some notation. Reflection $z \mapsto -\bar{z}$ interchanges the holes and preserves the outer circle, so

$$(35) \quad \pi C = \begin{pmatrix} d & -b \\ -b & d \end{pmatrix}.$$

Indeed, uniqueness for the Dirichlet problem gives $\omega_+(-\bar{z}) = \omega_-(z)$. Changing variables in the energy integral gives $C_{++} = C_{--}$, while $C_{+-} = C_{-+}$ follows from symmetry. The notation in (35) therefore means precisely

$$d := \pi C_{++} = \pi C_{--}, \quad b := -\pi C_{+-}, \quad \tau = -\pi C = \begin{pmatrix} -d & b \\ b & -d \end{pmatrix}.$$

Thus d and b are defined from the capacity matrix of this domain. Here $b > 0$: on either inner boundary the other harmonic measure vanishes, is strictly positive in the interior, and has strictly negative outward normal derivative by the Hopf lemma. Hence $C_{+-} < 0$. Positive definiteness of πC implies that its eigenvalues $d - b$ and $d + b$ are positive, so $d > b > 0$, as required in (16).

4.1. Comparison with the exact single-hole solutions. Put

$$(36) \quad W_{\pm}(z) = -\frac{1}{L} \log \left| \frac{z \mp c}{1 \mp cz} \right|.$$

This is the harmonic measure of the hole $K_{\pm}$ in the domain with that hole alone. It is 1 on its own inner circle, 0 on the unit circle, and has single-hole flux

$$(37) \quad C_* = \frac{2\pi}{L}, \quad d_* = \pi C_*.$$

To compute this constant, use $w = (z - c)/(1 - cz)$ for K_+ (and the corresponding map for K_-). The one-hole domain becomes $\rho < |w| < 1$ and the potential becomes $-\log |w|/L$. On $|w| = \rho$ the outward normal of the annulus is $-\partial_r$, hence

$$\partial_{\nu_w} \left(-\frac{\log r}{L} \right) = \frac{1}{\rho L}, \quad \int_{|w|=\rho} \frac{1}{\rho L} ds_w = \int_0^{2\pi} \frac{1}{\rho L} \rho d\theta = \frac{2\pi}{L}.$$

The flux is unchanged by a conformal coordinate: if $w = T(z)$ and $v(z) = \tilde{v}(w)$, then

$$\partial_{\nu_z} v = |T'(z)| \partial_{\nu_w} \tilde{v}, \quad ds_w = |T'(z)| ds_z, \quad \partial_{\nu_z} v ds_z = \partial_{\nu_w} \tilde{v} ds_w.$$

This proves (37) in the original coordinate as well.

For the interaction parameter define

$$(38) \quad a = \frac{2c}{1+c^2}, \quad \gamma = -\log a.$$

For c sufficiently close to 1, $a > \rho$. The automorphism $u = (z+c)/(1+cz)$ maps K_- to $\{|u| \leq \rho\}$, and in this coordinate

$$(39) \quad W_+(z(u)) = -\frac{1}{L} \log \left| \frac{u-a}{1-au} \right|.$$

For example, solving for z gives $z = (u-c)/(1-cu)$, and direct substitution yields

$$\frac{z-c}{1-cz} = \frac{(1+c^2)u-2c}{1+c^2-2cu} = \frac{u-a}{1-au},$$

which proves (39). On $u = \rho e^{i\theta}$, put $x = \cos \theta$. The squared modulus of this quotient is

$$F(x) = \frac{a^2 + \rho^2 - 2a\rho x}{1 + a^2\rho^2 - 2a\rho x}, \quad F'(x) = -\frac{2a\rho(1-a^2)(1-\rho^2)}{(1+a^2\rho^2-2a\rho x)^2} < 0.$$

Since $-\log \sqrt{F(x)}$ is increasing, its minimum occurs at $x = -1$ and its maximum at $x = 1$. The corresponding moduli are $(a+\rho)/(1+a\rho)$ and $(a-\rho)/(1-a\rho)$; all terms are positive because $\rho < a < 1$. Thus the minimum and maximum of W_+ on ∂K_- are

$$(40) \quad \alpha = \frac{1}{L} \log \frac{1+a\rho}{a+\rho}, \quad \beta = \frac{1}{L} \log \frac{1-a\rho}{a-\rho}.$$

The same numbers apply with $+$ and $-$ interchanged. Both are strictly positive. To expand them, regard α and β as functions of a and note

$$\begin{aligned} \alpha(1) &= \beta(1) = 0, \\ \alpha'(1) &= \frac{1}{L} \left(\frac{\rho}{1+\rho} - \frac{1}{1+\rho} \right) = -\frac{1-\rho}{L(1+\rho)}, \\ \beta'(1) &= \frac{1}{L} \left(-\frac{\rho}{1-\rho} - \frac{1}{1-\rho} \right) = -\frac{1+\rho}{L(1-\rho)}. \end{aligned}$$

Because $a = e^{-\gamma} = 1 - \gamma + O(\gamma^2)$, ordinary Taylor expansion at $a = 1$ gives

$$(41) \quad \alpha = \frac{1-\rho}{L(1+\rho)}\gamma + O(\gamma^2), \quad \beta = \frac{1+\rho}{L(1-\rho)}\gamma + O(\gamma^2).$$

Lemma 4.1. *Let $V_+ = W_+ - \omega_+$. Then*

$$(42) \quad \alpha\omega_- \leq V_+ \leq \beta\omega_- \quad \text{on } \Omega_c.$$

The analogous inequalities hold with the labels interchanged. Moreover,

$$(43) \quad \alpha d \leq b \leq \beta d,$$

$$(44) \quad \alpha b \leq d - d_* \leq \beta b.$$

In particular, when $\beta < 1$,

$$(45) \quad \frac{d_*}{1 - \alpha^2} \leq d \leq \frac{d_*}{1 - \beta^2}.$$

Thus $d = d_* + O(\gamma^2)$ and $b \asymp \gamma$.

Proof. The function V_+ is harmonic, vanishes on Γ_0 and ∂K_+ , and equals $W_+ \in [\alpha, \beta]$ on ∂K_- . The maximum principle proves (42).

To spell out the boundary terms, Green's second identity gives

$$0 = \int_{\Omega_c} (\omega_- \Delta V_+ - V_+ \Delta \omega_-) dA = \int_{\partial \Omega_c} (\omega_- \partial_\nu V_+ - V_+ \partial_\nu \omega_-) ds.$$

On Γ_0 and ∂K_+ both ω_- and V_+ vanish, so those two components contribute zero. On ∂K_- , their values are $\omega_- = 1$ and $V_+ = W_+$. Rearranging the remaining boundary integral gives

$$(46) \quad \int_{\partial K_-} \partial_\nu V_+ ds = \int_{\partial K_-} W_+ \partial_\nu \omega_- ds.$$

The flux of W_+ around K_- is zero because W_+ is harmonic through that disk. More explicitly, the outward normal of the disk is $-\nu$, and the divergence theorem gives

$$\int_{\partial K_-} \partial_\nu W_+ ds = - \int_{K_-} \Delta W_+ dA = 0.$$

Therefore

$$\int_{\partial K_-} \partial_\nu V_+ ds = \int_{\partial K_-} (\partial_\nu W_+ - \partial_\nu \omega_-) ds = 0 - C_{-+} = \frac{b}{\pi}.$$

On its own inner boundary $\partial_\nu \omega_- > 0$, with total flux C_{--} . Bounding W_+ between α and β in (46), and multiplying by π , proves (43).

On ∂K_+ all three functions in (42) vanish. If a differentiable function f is nonnegative in the domain and zero at a boundary point z , then

$$\partial_\nu f(z) = \lim_{h \downarrow 0} \frac{f(z) - f(z - h\nu)}{h} \leq 0.$$

Apply this to $V_+ - \alpha \omega_-$ and $\beta \omega_- - V_+$. It explains why differentiation in the outward normal reverses the inequalities:

$$\beta \partial_\nu \omega_- \leq \partial_\nu V_+ \leq \alpha \partial_\nu \omega_-.$$

The integral of the middle term is

$$\int_{\partial K_+} \partial_\nu V_+ ds = \int_{\partial K_+} \partial_\nu W_+ ds - \int_{\partial K_+} \partial_\nu \omega_+ ds = C_* - C_{++}.$$

In particular, C_* is the flux of W_+ itself; the correction V_+ has flux $C_* - C_{++}$. The integrated inequalities are

$$\beta C_{+-} \leq C_* - C_{++} \leq \alpha C_{+-}.$$

Since $C_{+-} = -b/\pi$ and $C_* - C_{++} = (d_* - d)/\pi$, multiplication by $-\pi$ yields $\alpha b \leq d - d_* \leq \beta b$, namely (44). Combining (43) and (44) gives $\alpha^2 d \leq d - d_* \leq \beta^2 d$, which is exactly (45). Finally (41) proves the stated orders and the positive lower bound implicit in $b \asymp \gamma$. $\square$

4.2. The leading interaction coefficient. The preceding estimates already suffice to prove opposite signs. The next refinement gives the explicit leading coefficient in Theorem 1.1.

Proposition 4.2. *As $c \rightarrow 1^-$,*

$$(47) \quad b = \frac{2\pi^2}{L^2} \gamma + O(\gamma^3).$$

Consequently, with $c = 1 - \varepsilon$,

$$(48) \quad d = d_* + O(\varepsilon^4), \quad b = \frac{\pi^2}{L^2} \varepsilon^2 + O(\varepsilon^3).$$

Proof. Work in the u coordinate of (39). Choose a fixed r_0 with $\rho < r_0 < 1$. For c near 1 the other hole is outside $|u| \leq r_0$. On this fixed disk the cross potential in (39), and each fixed finite number of its derivatives, is $O(\gamma)$: the logarithm is smooth in (u, a) for $|u| \leq r_0$ and a near 1, and it vanishes at $a = 1$.

Use (42) with the labels interchanged:

$$0 \leq V_- := W_- - \omega_- \leq \beta \omega_+ \leq \beta W_+.$$

Thus $V_- = O(\gamma^2)$ on the fixed collar $\rho \leq |u| \leq r_0$, and $V_- = 0$ on $|u| = \rho$. Choose M independent of c so that $V_- \leq M\gamma^2$ on $|u| = r_0$. The radial harmonic function

$$B(u) = M\gamma^2 \frac{\log(|u|/\rho)}{\log(r_0/\rho)}$$

vanishes on $|u| = \rho$ and equals $M\gamma^2$ on $|u| = r_0$. The maximum principle on this fixed annulus gives $0 \leq V_- \leq B$. Using the outward normal $\nu_u = -\partial_r$ at the inner circle and the preceding boundary derivative rule, we obtain

$$-\frac{M\gamma^2}{\rho \log(r_0/\rho)} = \partial_{\nu_u} B \leq \partial_{\nu_u} V_- \leq 0.$$

Thus the correction to the normal derivative is $O(\gamma^2)$ uniformly along the circle. Since $W_-(z(u)) = -\log|u|/L$, the transformed harmonic measure $\tilde{\omega}_-(u) = \omega_-(z(u))$ satisfies

$$(49) \quad \partial_{\nu_u} \tilde{\omega}_-(u) = \frac{1}{\rho L} + O(\gamma^2), \quad |u| = \rho.$$

Flux measure is conformally invariant, so (46) becomes

$$\frac{b}{\pi} = \int_{|u|=\rho} \left(-\frac{1}{L} \log \left| \frac{u-a}{1-au} \right| \right) \partial_{\nu_u} \tilde{\omega}_-(u) ds_u.$$

The logarithmic factor is $O(\gamma)$ on this circle. Its mean is exactly γ/L : it is harmonic on $|u| \leq \rho$ and its value at 0 is $-\log a/L$. Writing this potential as $F(u)$ and the normal derivative error as $E(u)$, we have $\sup |F| = O(\gamma)$ and $\sup |E| = O(\gamma^2)$. The mean-value property says

$$\int_{|u|=\rho} F(u) ds_u = 2\pi\rho F(0) = \frac{2\pi\rho\gamma}{L}, \quad \left| \int_{|u|=\rho} F(u)E(u) ds_u \right| = O(\gamma^3).$$

Inserting (49) therefore gives

$$\frac{b}{\pi} = \frac{1}{\rho L} \frac{2\pi\rho\gamma}{L} + O(\gamma^3),$$

which proves (47). Lastly,

$$\gamma = \log \frac{1+c^2}{2c} = \log \left(1 + \frac{\varepsilon^2}{2(1-\varepsilon)} \right) = \frac{1}{2}\varepsilon^2 + O(\varepsilon^3).$$

This and Lemma 4.1 prove (48). $\square$

4.3. A pole whose displacement is smaller than the squared coupling.

Proposition 4.3. *At $p_\varepsilon = i(1 - \varepsilon^5)$,*

$$(50) \quad \omega_+(p_\varepsilon) = \omega_-(p_\varepsilon) =: h_\varepsilon,$$

and

$$(51) \quad \delta_\varepsilon := 2\pi h_\varepsilon = \frac{2\pi}{L}\varepsilon^6 + O(\varepsilon^7).$$

In particular $\delta_\varepsilon/b_\varepsilon^2 \rightarrow 0$.

Proof. The reflection $z \mapsto -\bar{z}$ fixes p_ε , so the two measures are equal. The single-hole potentials are also equal there; denote their common value by w_ε . Evaluating (42) at this point gives

$$\alpha h_\varepsilon \leq w_\varepsilon - h_\varepsilon \leq \beta h_\varepsilon, \quad \frac{w_\varepsilon}{1+\beta} \leq h_\varepsilon \leq \frac{w_\varepsilon}{1+\alpha}.$$

Since $\alpha, \beta = O(\varepsilon^2)$, it follows that $h_\varepsilon = w_\varepsilon(1 + O(\varepsilon^2))$.

For $y = 1 - \varepsilon^5$ and $c = 1 - \varepsilon$, the value is explicit:

$$(52) \quad \begin{aligned} w_\varepsilon &= -\frac{1}{2L} \log \frac{y^2 + c^2}{1 + c^2 y^2} \\ &= -\frac{1}{2L} \log \left(1 - \frac{(1-c^2)(1-y^2)}{1 + c^2 y^2} \right). \end{aligned}$$

The identity in the second line follows from $1 + c^2 y^2 - (y^2 + c^2) = (1 - c^2)(1 - y^2)$. Its small fraction can be computed term by term:

$$\begin{aligned} 1 - c^2 &= 2\varepsilon - \varepsilon^2, & 1 - y^2 &= 2\varepsilon^5 - \varepsilon^{10}, \\ (1 - c^2)(1 - y^2) &= 4\varepsilon^6 + O(\varepsilon^7), & 1 + c^2 y^2 &= 2 + O(\varepsilon). \end{aligned}$$

Thus the fraction is $2\varepsilon^6 + O(\varepsilon^7)$. Since $-\log(1 - x) = x + O(x^2)$,

$$w_\varepsilon = \frac{1}{2L}(2\varepsilon^6 + O(\varepsilon^7) + O(\varepsilon^{12})) = \frac{\varepsilon^6}{L} + O(\varepsilon^7).$$

Multiplication by $1 + O(\varepsilon^2)$ leaves this leading term and its displayed error unchanged. Hence $h_\varepsilon = \varepsilon^6/L + O(\varepsilon^7)$, and multiplication by 2π proves (51). Finally Proposition 4.2 gives

$$b_\varepsilon^2 = \frac{\pi^4}{L^4}\varepsilon^4 + O(\varepsilon^5), \quad \frac{\delta_\varepsilon}{b_\varepsilon^2} = \frac{2L^3}{\pi^3}\varepsilon^2 + O(\varepsilon^3) \longrightarrow 0.$$

□

5. IDENTIFICATION WITH THE GREEN-FUNCTION PARAMETER

The following is the specific external extremal theorem used in the proof.

Theorem 5.1 (Yamada, [Yam99, Theorem 2.1]). *Let Ω be a regular planar domain, and use the normalized symmetric homology basis and theta convention of Section 3. For $p \in \Omega$, evaluate $p - \phi(p)$ along a symmetric path. On $i\mathbb{R}^g/(2\pi i\mathbb{Z}^g)$ the positive function*

$$(53) \quad e \longmapsto \frac{\theta_\tau(p - \phi(p) + e)}{\theta_\tau(e)}$$

has its unique maximum at $e = e_0(p)$. Here $e_0(p)$ is the Green critical-divisor parameter in (4), with multiplicities. In particular this characterization does not require the critical points to be simple.

We use this theorem with its uniqueness assertion, not a converse inference from the scalar Saitoh inequality. The sign question for the individual matrix $H_\tau(e_0(p))$ is not part of the quoted theorem.

Proof of Theorem 1.1. For Ω_ε , equations (33) and (35) identify its theta function with $\Theta_{d_\varepsilon, b_\varepsilon}$. The parameters satisfy

$$d_\varepsilon = d_* + O(\varepsilon^4), \quad b_\varepsilon = \frac{\pi^2}{L^2}\varepsilon^2 + O(\varepsilon^3) > 0.$$

By Lemma 3.1 and Proposition 4.3,

$$p_\varepsilon - \phi(p_\varepsilon) = -i\delta_\varepsilon(1, 1), \quad \delta_\varepsilon = \frac{2\pi}{L}\varepsilon^6 + O(\varepsilon^7) > 0.$$

Taking the logarithm of (53), with $e = it$, gives exactly $R_{d_\varepsilon, b_\varepsilon, \delta_\varepsilon}(t)$ from (25). Both Theorem 5.1 and Proposition 2.4 characterize its unique global maximum. Hence

$$(54) \quad e_0(p_\varepsilon) = iv_{d_\varepsilon, b_\varepsilon, \delta_\varepsilon}(1, 1) \pmod{2\pi i\mathbb{Z}^2}.$$

The Hessian is $2\pi i\mathbb{Z}^2$ -periodic, so this ambiguity is irrelevant. This step uses the full theta function of the geometric period matrix, not its asymptotic approximation: the estimates of d_ε and b_ε enter only after the exact identification of the maximizer.

Smooth dependence in Lemma 2.1 yields $\kappa_{d_\varepsilon} = \kappa_* + O(\varepsilon^4)$. Substitution into (27) gives

$$\lambda_{\parallel} = -\frac{\kappa_*}{2} \frac{2\pi}{L} \varepsilon^6 + O(\varepsilon^7) + O(\varepsilon^8) + O(\varepsilon^{12}),$$

Here the $O(\varepsilon^7)$ term includes the error in δ_ε , the term $|b_\varepsilon|\delta_\varepsilon$ is $O(\varepsilon^8)$, and $\delta_\varepsilon^2 = O(\varepsilon^{12})$. The error in κ_{d_ε} contributes only $O(\varepsilon^{10})$. All are absorbed into $O(\varepsilon^7)$, proving (12). For the transverse direction, $|b_\varepsilon|^3 + \delta_\varepsilon = O(\varepsilon^6)$, and (28) gives

$$\lambda_{\perp} = \kappa_*^2 \left(\frac{\pi^2}{L^2} \varepsilon^2 + O(\varepsilon^3) \right)^2 + O(\varepsilon^6) = \frac{\pi^4 \kappa_*^2}{L^4} \varepsilon^4 + O(\varepsilon^5).$$

This is (13). The constants in both leading terms are strictly positive because $L > 0$ and $\kappa_* > 0$. $\square$

Corollary 5.2. *For c sufficiently close to 1, fix the domain Ω_c with the two holes (7). There exists $y_0(c) < 1$ such that $H_{\tau_c}(e_0(iy))$ is indefinite for every $y_0(c) < y < 1$.*

Proof. First choose c sufficiently close to 1 that $b_c > 0$ is small and d_c belongs to the fixed interval I . Proposition 2.3 then gives a strictly positive transverse eigenvalue at $\delta = 0$, because its leading coefficient is $\kappa_{d_c}^2 > 0$. For the longitudinal eigenvalue, (27) can be written as

$$\lambda_{\parallel}(d_c, b_c, \delta) = \delta \left(-\frac{\kappa_{d_c}}{2} + O(b_c) + O(\delta) \right).$$

By choosing c closer to 1 if necessary, the $O(b_c)$ term is smaller in absolute value than $\kappa_{d_c}/4$. Now fix this c . For every sufficiently small $\delta > 0$ the bracket remains negative. Continuity keeps the transverse eigenvalue positive for all sufficiently small δ . Along the imaginary axis the common harmonic measure is positive inside and tends to zero at i . Theorem 5.1 identifies these maxima with $e_0(iy)$ just as above. $\square$

6. THE DEFINITENESS PROBLEM AND THE SCALAR INEQUALITY

The example disproves the universal negative-definiteness assertion (5) already in genus two. It does not classify the signature of $H_\tau(e)$ at every point of every Jacobian. In particular, it does not contradict positivity at $e = 0$: directly from (1), $H_\tau(0)$ is the covariance matrix of a strictly positive distribution on $\mathbb{Z}^g$, and is positive definite.

The distinction between a scalar kernel inequality and matrix definiteness is essential here. The extremal characterization used in Theorem 5.1 controls the variation of a *ratio* of theta functions. The second derivative of its

logarithm compares the Hessians at two points; it does not prescribe the signature at either point separately.

Yamada’s relative comparison [Yam99, Theorem 3.2] concerns the difference of Hessians at two translated points. Such a difference can be positive definite even when one of the individual Hessians is indefinite. Likewise, the scalar Saitoh inequality proved in [Gua19] does not imply that every real direction in the theta parameter space is negative. The counterexample addresses precisely this distinction.

APPENDIX A. NUMERICAL ILLUSTRATIONS

We give two formulas for evaluating the Hessian and illustrate the asymptotic regimes in Theorem 1.1 and Proposition 2.3. The accompanying NumPy script implements these formulas and a circular-multipole approximation of harmonic measures. These floating-point computations are supplementary: the existence assertion in Theorem 1.1 follows from the analytic estimates in Sections 2–5. Boundary collocation residuals are not used as rigorous error bounds.

A.1. Two independent formulas for the theta Hessian. Write

$$A = \begin{pmatrix} d & -b \\ -b & d \end{pmatrix}, \quad w_n(t) = e^{-n^T A n/2 + i n \cdot t}, \quad Z(t) = \sum_{n \in \mathbb{Z}^2} w_n(t).$$

Direct differentiation of the Fourier series gives

$$(55) \quad H_{jk}(it) = \frac{\sum_n n_j n_k w_n(t)}{Z(t)} - \frac{(\sum_n n_j w_n(t))(\sum_n n_k w_n(t))}{Z(t)^2}.$$

Here the products do not use complex conjugation. A second formula follows from (18). Give $X = 2\pi k$ probability proportional to $\exp[-(t - 2\pi k)^T A^{-1}(t - 2\pi k)/2]$. Then

$$(56) \quad H(it) = A^{-1} - A^{-1} \text{Cov}(X) A^{-1}.$$

For the differentiation behind this identity, let $\mu_k(t)$ be the normalized Gaussian weights, $X_k = 2\pi k$, and $m(t) = \sum_k \mu_k(t) X_k$. The prefactor in (18) is independent of t , so

$$\nabla \mathcal{L}(t) = -A^{-1}(t - m(t)), \quad \nabla \log \mu_k(t) = A^{-1}(X_k - m(t)).$$

Consequently $D_t m = \text{Cov}(X) A^{-1}$, and differentiating once more yields

$$D_t^2 \mathcal{L} = -A^{-1} + A^{-1} \text{Cov}(X) A^{-1}.$$

Changing the sign to pass to the holomorphic Hessian proves (56). The script compares (55) and (56), rather than inferring signs from finite differences of a logarithm.

For $d = 4$, the scalar constants are approximately

$$t_d = 1.8354437625, \quad \kappa_d = 0.30050166, \quad \kappa_d^2 = 0.09030125.$$

At the maximum of $S_{d,b}$, the values in Table 1 illustrate the convergence $\lambda_{\perp}/b^2 \rightarrow \kappa_d^2$.

TABLE 1. The transverse eigenvalue at the auxiliary maximum, with $d = 4$.

	b	$\lambda_{\perp}$	$\lambda_{\perp}/b^2$
0.100	8.8096501×10^{-4}		0.08809650
0.050	2.2312796×10^{-4}		0.08925118
0.020	3.5957394×10^{-5}		0.08989348
0.010	9.0099383×10^{-6}		0.09009938
0.005	2.2550204×10^{-6}		0.09020082

A.2. Harmonic measure by circular multipoles. For numerical evaluation only, write the two Euclidean holes as $|z - c_j| \leq r_j$. Harmonic basis functions vanishing on the unit circle are

$$\begin{aligned}
B_{j,0}(z) &= \log \left| \frac{z - c_j}{1 - \bar{c}_j z} \right|, \\
B_{j,n}^R(z) &= \operatorname{Re} \left[\left(\frac{r_j}{z - c_j} \right)^n - \left(\frac{r_j z}{1 - \bar{c}_j z} \right)^n \right], \\
B_{j,n}^I(z) &= \operatorname{Im} \left[\left(\frac{r_j}{z - c_j} \right)^n + \left(\frac{r_j z}{1 - \bar{c}_j z} \right)^n \right], \quad n \geq 1.
\end{aligned}$$

Boundary collocation determines coefficients approximating each ω_k . The logarithmic coefficient a_{jk} determines the flux exactly for that finite harmonic approximation, since the multipoles and image terms have zero net flux:

$$C_{jk}^{\text{approx}} = -2\pi a_{jk}, \quad A_{jk}^{\text{approx}} = -2\pi^2 a_{jk}.$$

Increasing both the multipole order and the number of boundary sample points tests numerical stability. The exact bounds (43) and (45) provide additional checks independent of collocation.

For the illustrative choice $\rho = e^{-\pi^2/2}$, $c = 0.85$, and $p = 0.99999i$, representative numerical values are

$$A \approx \begin{pmatrix} 4.0000284261 & -0.0106578265 \\ -0.0106578265 & 4.0000284261 \end{pmatrix},$$

$$\omega_+(p) = \omega_-(p) \approx 3.25596257 \times 10^{-7}.$$

The eigenvalues at the numerically located maximum are approximately

$$\lambda_{\parallel} \approx -3.074 \times 10^{-7}, \quad \lambda_{\perp} \approx 9.925 \times 10^{-6}.$$

These particular decimal parameters are a numerical illustration, not a separately certified explicit finite-parameter counterexample. Theorem 1.1 establishes the counterexample family by analytic asymptotics.

A.3. Avoiding cancellation for small displacements. On the diagonal let $J(u) = \mathcal{L}_{d,b}(u, u)$. The critical-point equation for the finite ratio is $J'(v - \delta) = J'(v)$. Direct subtraction loses precision when δ is tiny. Since $J'' = -2\Lambda_{\parallel}$, the same equation is

$$\int_{-1/2}^{1/2} \Lambda_{\parallel}(w + \delta s) ds = 0, \quad v = w + \delta/2.$$

This equation can be solved using Gaussian quadrature and a bracketed root finder near $u_{d,b}$. At an approximate root w , the expression

$$\Lambda_{\parallel}(w + \delta/2) - \int_{-1/2}^{1/2} \Lambda_{\parallel}(w + \delta s) ds$$

reduces the effect of a residual common to the averaged and endpoint curvatures. At the exact root it equals the longitudinal eigenvalue. The accompanying calculation records both this value and the uncorrected endpoint curvature, compares two quadrature orders, and compares (55) with (56).

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DECLARATION ON THE USE OF ARTIFICIAL INTELLIGENCE

The author chose the problem, suggested searching for a counterexample, and supplied a related construction. The new construction, proofs, computations, exposition, and figures were developed with OpenAI Codex (GPT-6). The author has completed a self-review of the mathematical arguments and computations.

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